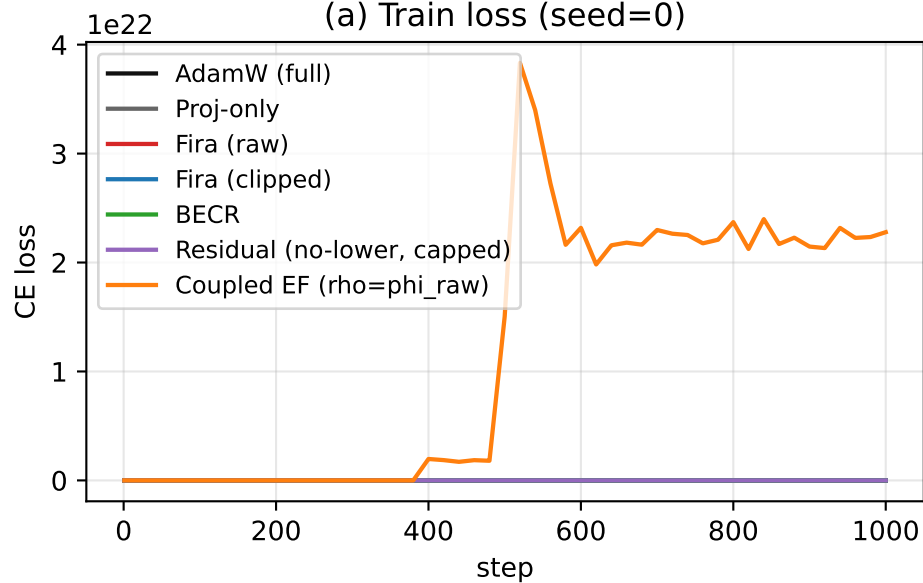
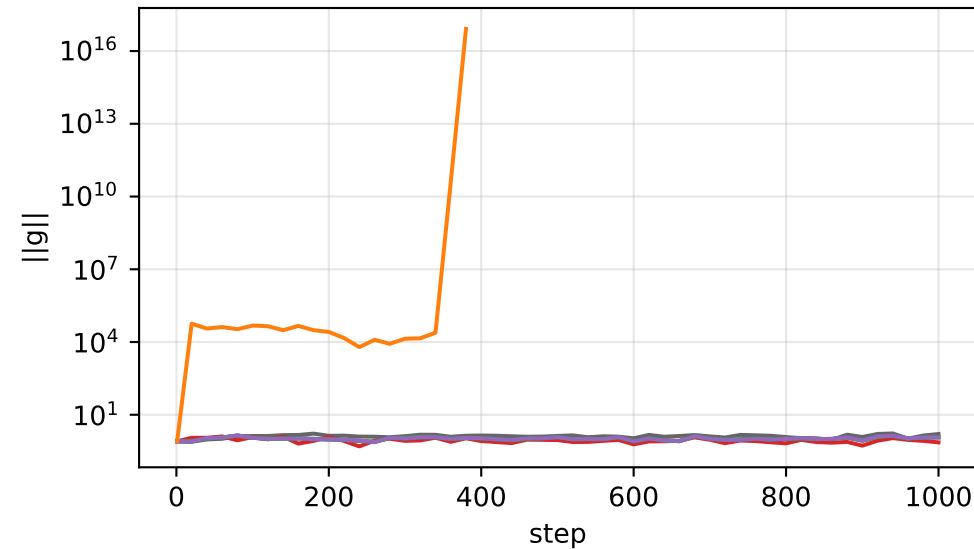


MNIST MLP (mechanism diagnostic, 1000 steps)

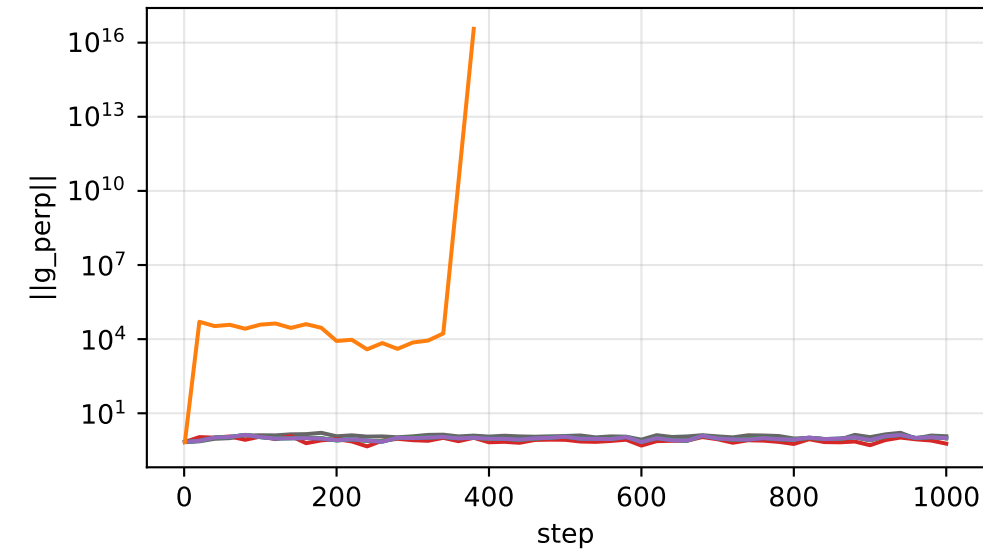
(a) Train loss (seed=0)



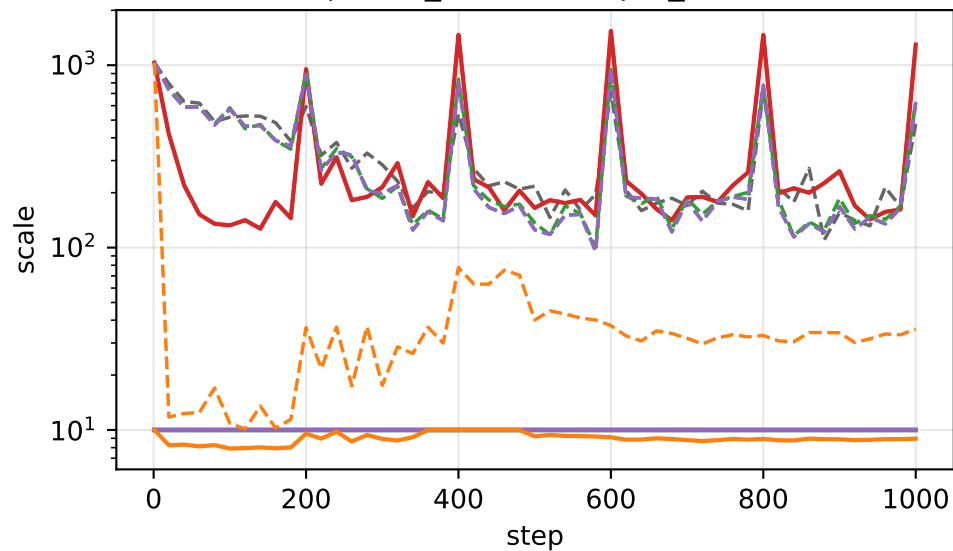
(b) $\|g\|$ (logged, diagnostic)



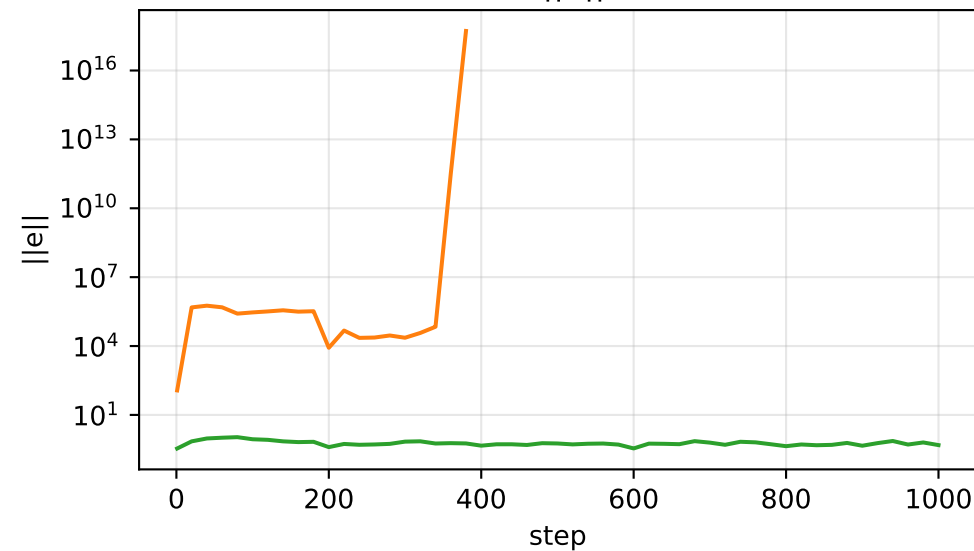
(c) $\|g_{\text{perp}}\|$ (logged, diagnostic)



(d) Scale p95: s_t (solid) vs ϕ_{raw} (dashed)



(e) Residual norm $\|e\|$ (BECR variants)



(f) Final test accuracy (mean $\pm$ std over seeds)

